Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 19: absolute error cannot be negative.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Calculate MAE MSE and RMSE. Try the recall before opening notes.

Your learning target

I can calculate MAE MSE and **RMSE**.

I can explain how squaring changes sensitivity to large errors.

I can match the **loss** and **unit**s to the task.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

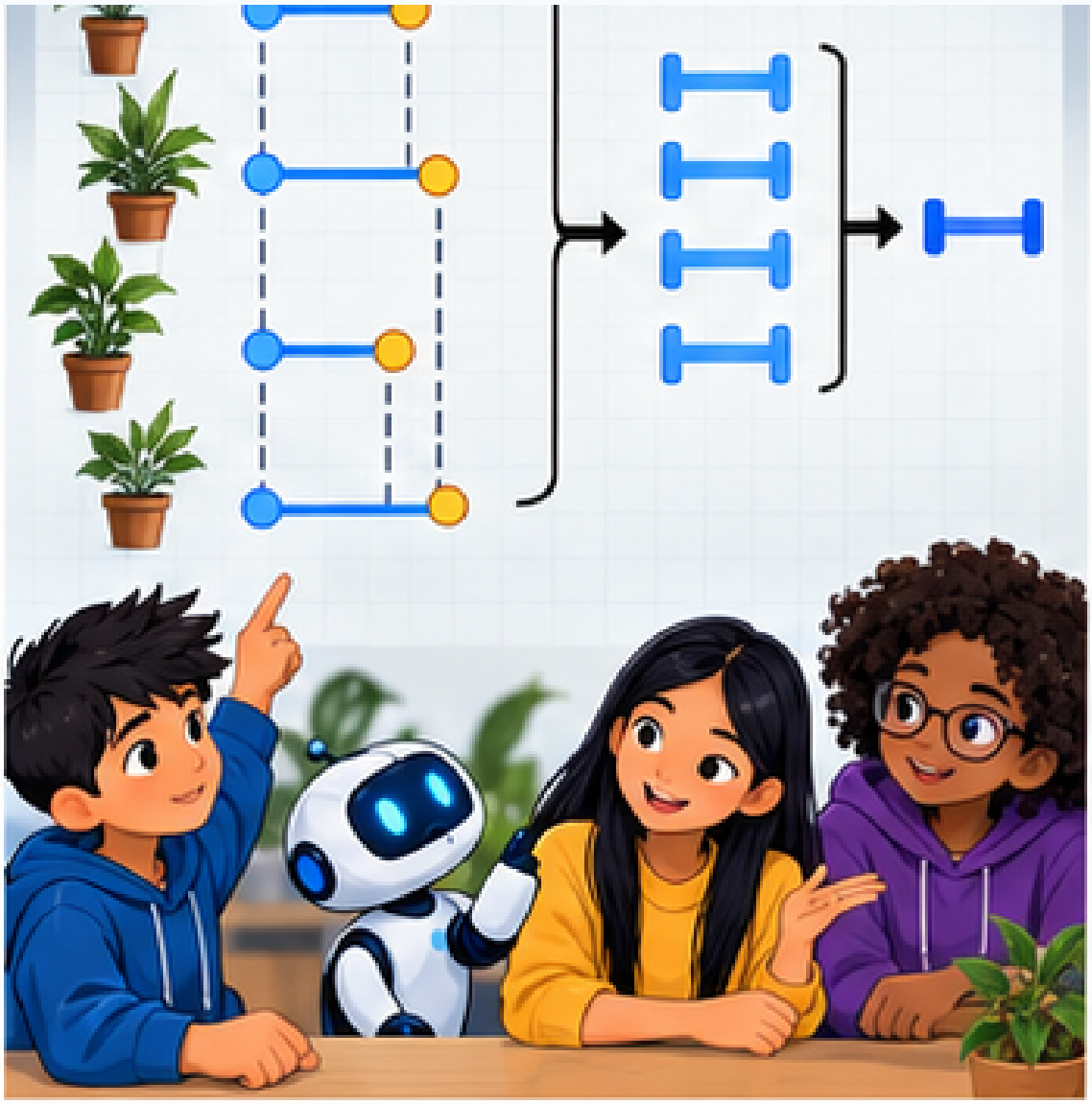
Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Absolute error measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.



Your target

LEARNING TARGET Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Builder name and experiment date:

Readiness for the connected explanation

Which constant minimizes **training** squared error?

This week's end goal

By the end, I can explain **absolute error** and mae using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

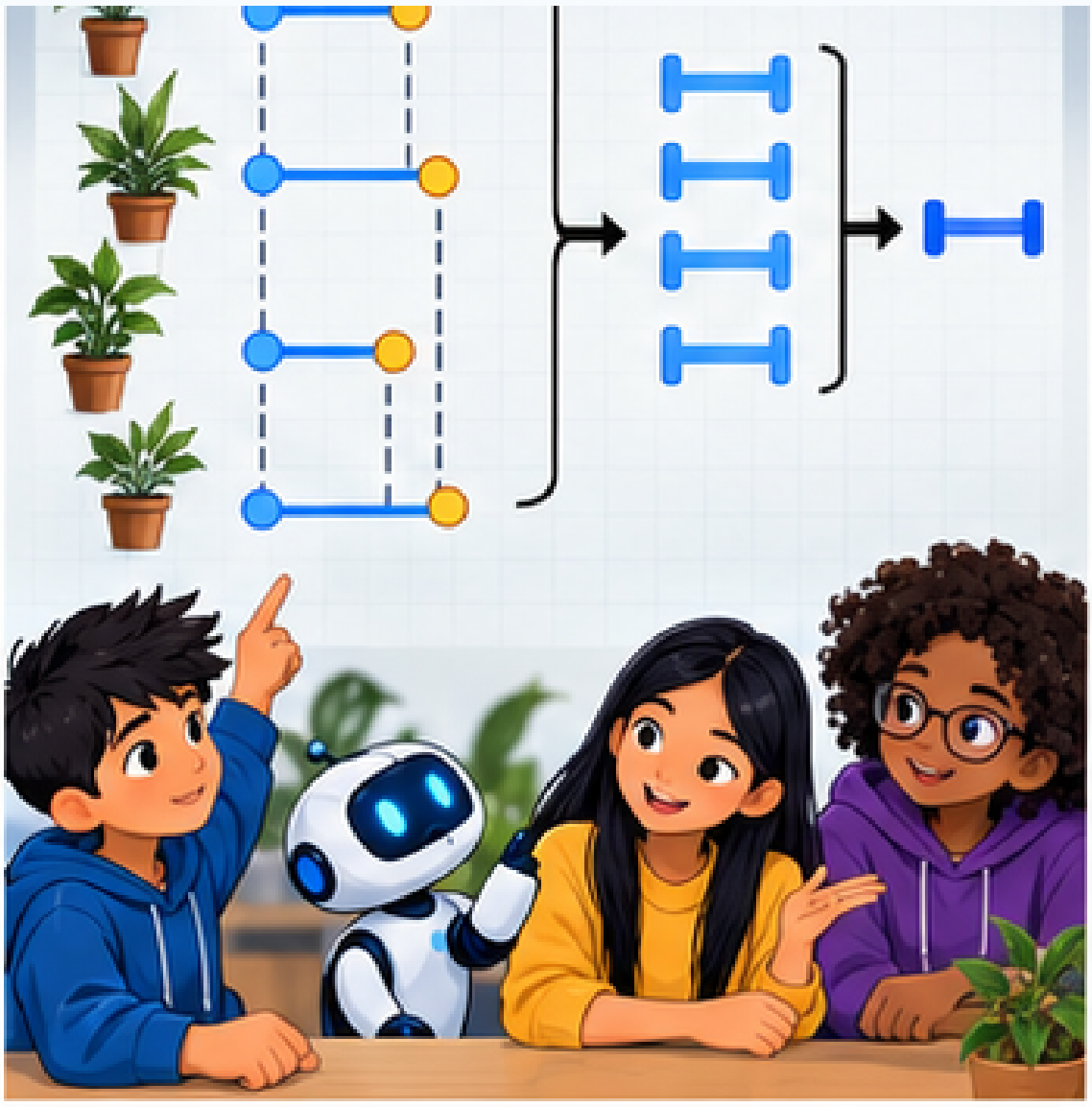
Notice the evidence

Investigate the supplied Absolute Error and MAE case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **absolute error** and mae to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A **loss** summarizes error magnitude without positive and negative **cancellation**. MAE averages absolute residuals. MSE averages squared residuals and penalizes large magnitudes more strongly.

RMSE takes the square root of MSE so it returns to the target **unit**. These are different questions, not interchangeable labels.

Use fictional classroom data throughout this investigation. Keep the task, **unit**s, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines absolute and squared **loss**.

STUDENT LESSON · 3 OF 18

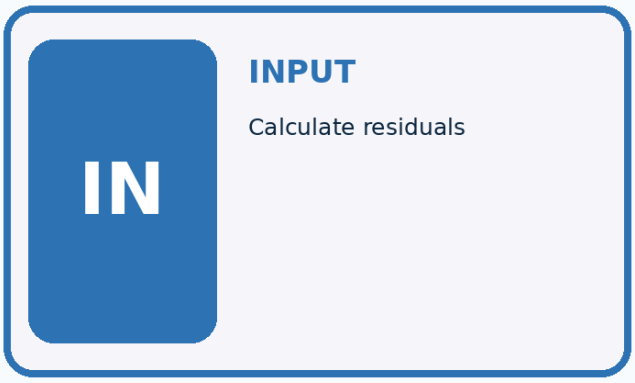
See how it works

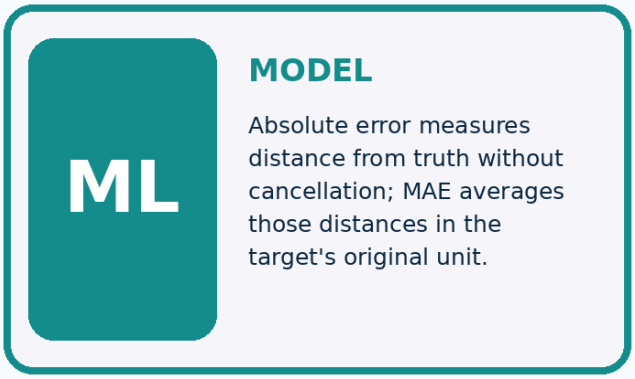
> Follow the mechanism for Absolute Error and MAE. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Calculate residuals, **absolute error**s, and mean **absolute error**.

MODEL IDEA **Absolute error** measures distance from truth without **cancellation**; MAE averages those distances in the target's original **unit**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17. Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Residuals [-1, 2,-3] cm have **absolute error**s [1, 2, 3], so MAE=6/3=2 cm. Squared errors [1, 4, 9] sum to 14, so MSE=14/3 square cm and **RMSE**=sqrt(14/3), about 2.16 cm. Keep the square and the average in the correct order.

```
from math import sqrt
residuals = [-1, 2, -3]
mae = sum(abs(r) for r in residuals) / len(residuals)
mse = sum(r * r for r in residuals) / len(residuals)
print(mae, mse, sqrt(mse))
# About 2.0, 4.67, 2.16.
# MAE and RMSE: cm. MSE: square cm.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

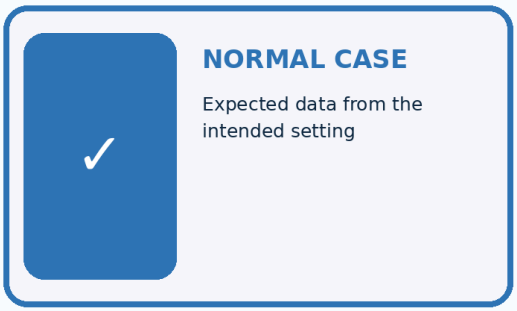
Compare the cases

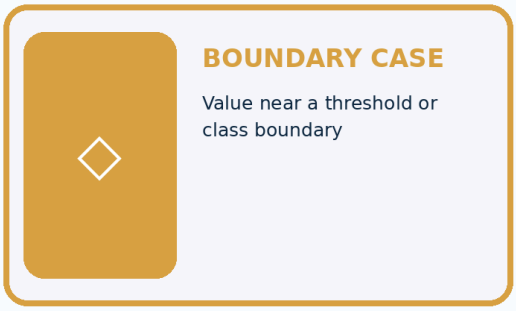
> Compare the supplied cases for Absolute Error and MAE. Identify the detail that changes the conclusion.

EVALUATION DESIGN

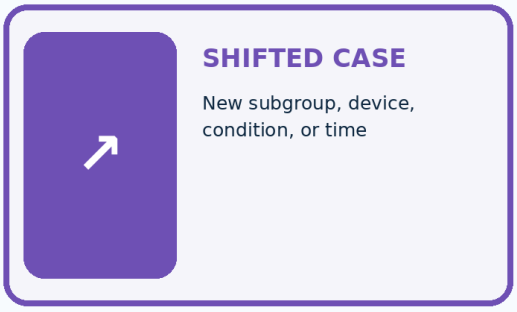
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Convert every residual to millimetres.

Predicted result

Prediction: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Why the result follows

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Absolute Error and MAE to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
ABSOLUTE ERROR
absolute value of actual minus predicted
2
MAE
mean of absolute errors
3
UNIT
measurement scale retained by MAE
4
CANCELLATION
positive and negative errors offset

Term	Precise meaning	My ML example
absolute error	absolute value of actual minus predicted	
MAE	mean of absolute errors	
unit	measurement scale retained by MAE	
cancellation	positive and negative errors offset	

MISCONCEPTION TO REPAIR Average signed residual is a safe replacement for average **absolute error**.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
MAE	Mean absolute error
MSE	Mean squared error
RMSE	Square root of mean squared error
Loss	A numerical penalty for prediction errors

STUDENT LESSON · 6 OF 18

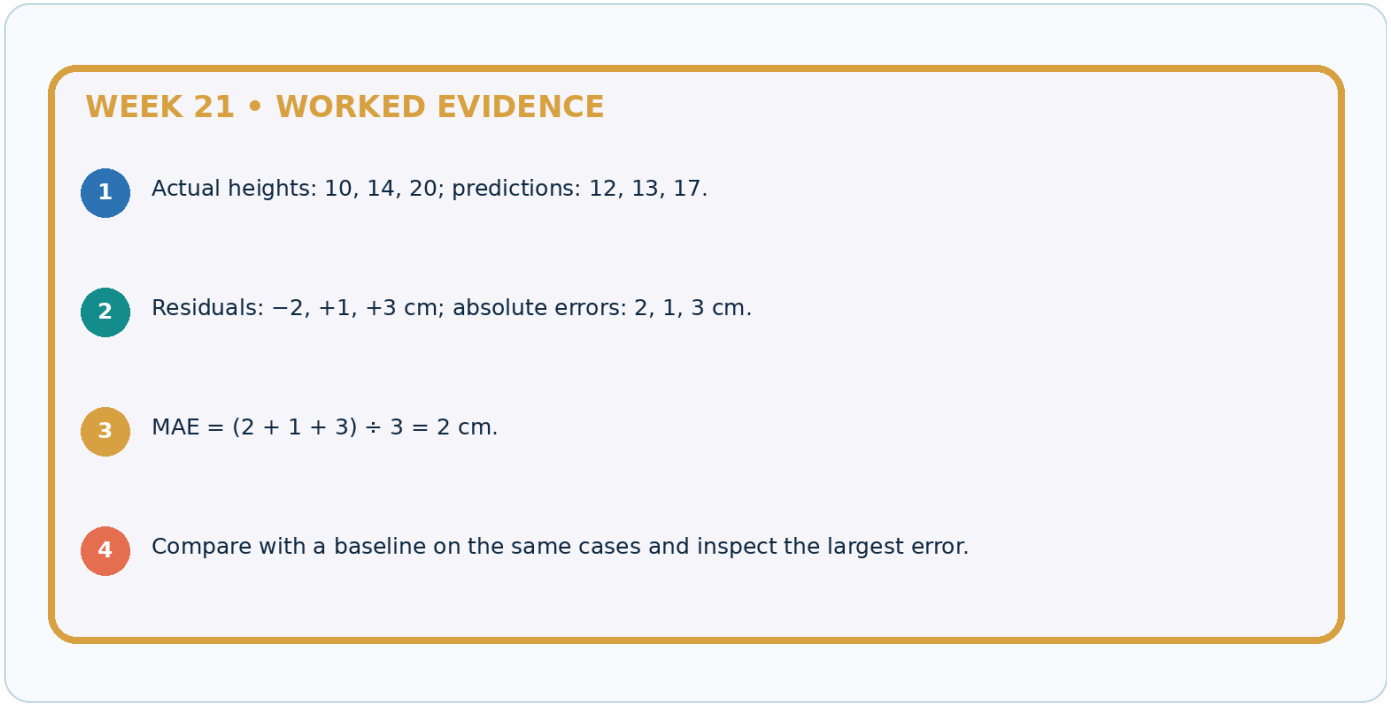
Follow a worked example

> Follow the worked trace for Absolute Error and MAE, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



Step 1

STEP 1 Actual heights: 10, 14, 20; **prediction**s: 12, 13, 17.

Step 2

STEP 2 Residuals: −2, +1, +3 cm; **absolute error**s: 2, 1, 3 cm.

Step 3

STEP 3 MAE = (2 + 1 + 3) ÷ 3 = 2 cm.

Step 4

STEP 4 Compare with a baseline on the same cases and inspect the largest error.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Absolute and squared loss at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Residuals	minus 1 plus 2 minus 3	Signed differences
Absolute values	1 2 3	MAE 2 cm
Squares	1 4 9	MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

Give MSE and RMSE with units. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Absolute Error and MAE

Compare absolute and squared error without losing track of **unit**s.

The facts we will use

Supplied facts

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/21

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Absolute errors

Absolute values1, 2, 3 give MAE6/3=2 cm.



2. Squared errors

Squares1, 4, 9 give MSE14/3 cm2, approximately = 4.67 cm2.



3. Root squared error

RMSE=sqrt(14/3), approximately2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Absolute errors	Absolute values1, 2, 3 give MAE6/3=2 cm.
2. Squared errors	Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm².
3. Root squared error	RMSE=sqrt(14/3), approximately2.16 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Keep the stated limit

Read the labelled picture: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Convert every residual to millimetres. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Absolute Error and MAE claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Average signed residual is a safe replacement for average absolute error. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Average signed residual is a safe replacement for average **absolute error**.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Choose and document the comparison metric before reviewing final-test results. Neither MAE nor MSE reveals all subgroup patterns or proves safety. Do not remove a large valid error solely because it worsens squared **loss**.

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

STUDENT LESSON · 8 OF 18

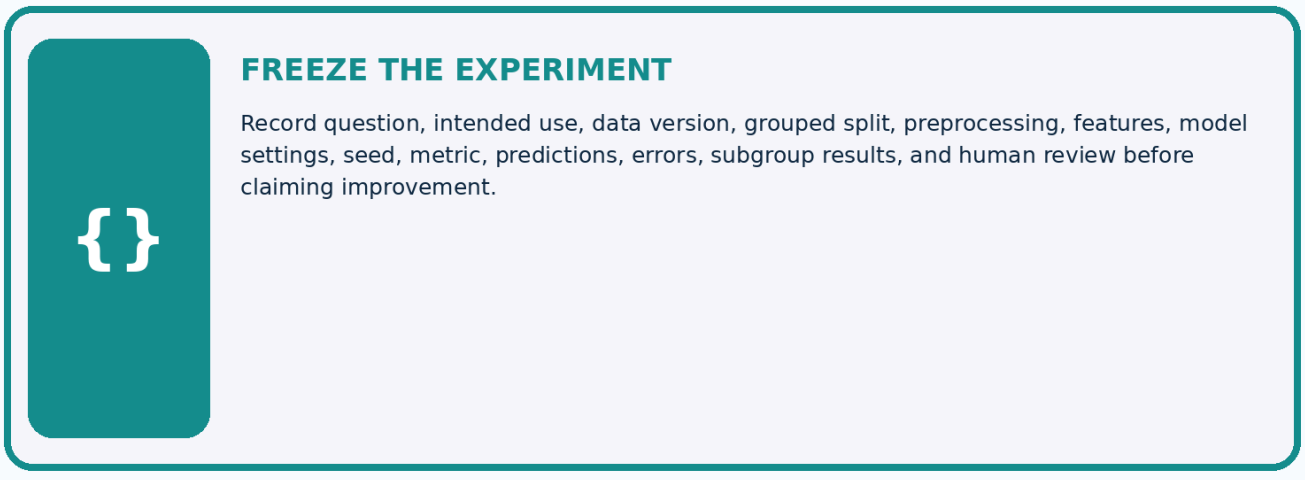
Practise together

Practise Absolute Error and MAE with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + **unit**s + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Changing cm to mm multiplies residuals and MAE/**RMSE** by 10, but MSE by 100. A **loss** decrease across reports is not meaningful unless **unit**s, evaluated rows and **loss** definitions match.

Retain the per-record residuals for investigation.

Practise with less help: Compute squared loss

Use workbook W2 for the connected example “Absolute and squared **loss**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> ---
>
> ---
>
> ---
>
> ---

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Absolute Error and MAE. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

MAE Lab: calculate each row, verify units and denominator, then compare model and baseline on identical examples.

1. Write the hypothesis and expected **evidence** before running.

2. MAE Lab: calculate each row, verify **unit**s and denominator, then compare model and baseline on identical examples.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

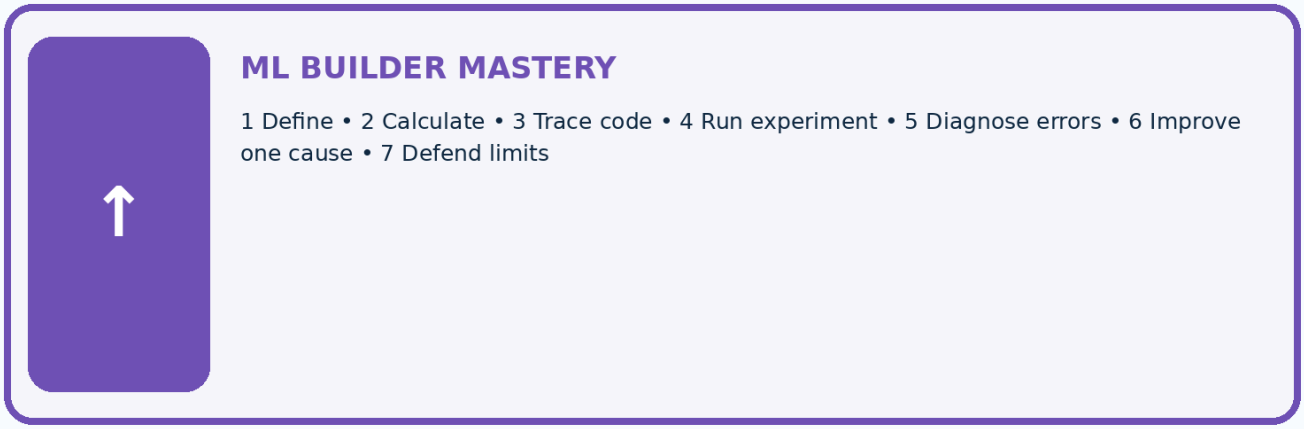
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Absolute Error and MAE. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **absolute error**, MAE, **unit**, **cancellation**.

CALCULATE Reproduce the worked **evidence** with **unit**s and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Average signed residual is a safe replacement for average **absolute error**.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

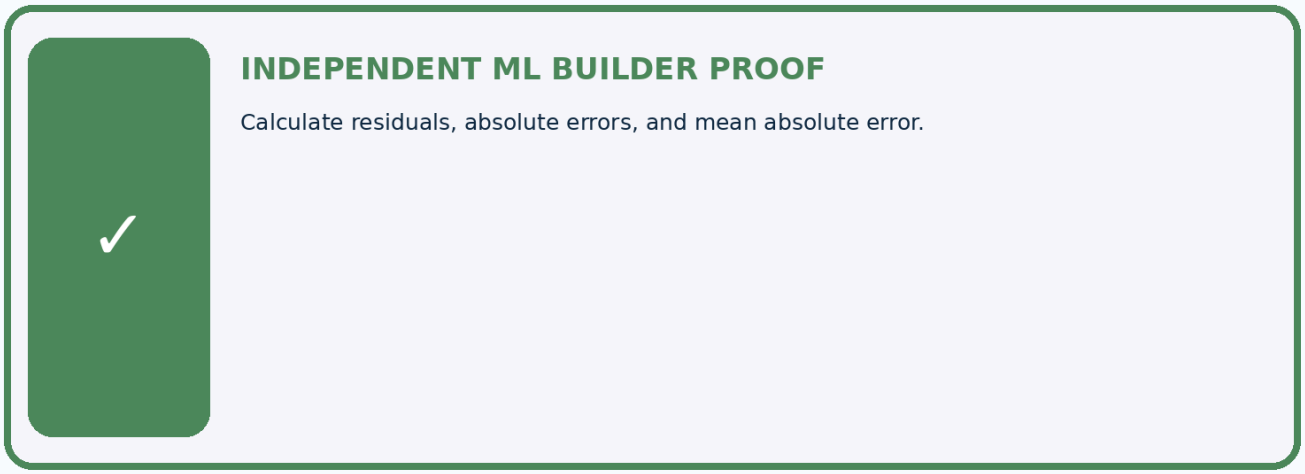
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Calculate residuals, **absolute error**s, and mean **absolute error**.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Train a Regression Model.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Absolute Error and MAE case for supported practice. Repeated examples do not become fresh assessments.

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Actual=[3,5,9], predicted=[2,7,6] cm.

The question

Calculate MAE.

Worked explanation

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Absolute errors are measured before averaging

Evidence or stage	Value or result
Case 1, actual 3, predicted 2	1
Case 2, actual 5, predicted 7	2
Case 3, actual 9, predicted 6	3

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Absolute errors=[1,2,3]; mean=2 cm. Do not cancel signed errors before taking absolute values.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Absolute Error and MAE, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compare absolute and squared error without losing track of **unit**s.

By the end, I can	How I will show it
Calculate MAE MSE and RMSE	Explain the deciding step, show a trace or test, and state one limit.
Explain how squaring changes sensitivity to large errors	Explain the deciding step, show a trace or test, and state one limit.
Match the loss and units to the task	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Absolute Error and MAE without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate MAE MSE and **RMSE**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare absolute and squared error without losing track of **unit**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **RMSE**=sqrt(14/3), approximately2.16 cm.

1. **Absolute error**s

2. Root squared error

3. Squared errors

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Root squared error **RMSE**=sqrt(14/3), approximately2.16 cm. Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Convert every residual to millimetres. Which conclusion is supported by the supplied facts?

1. MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Repair a missing building block

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

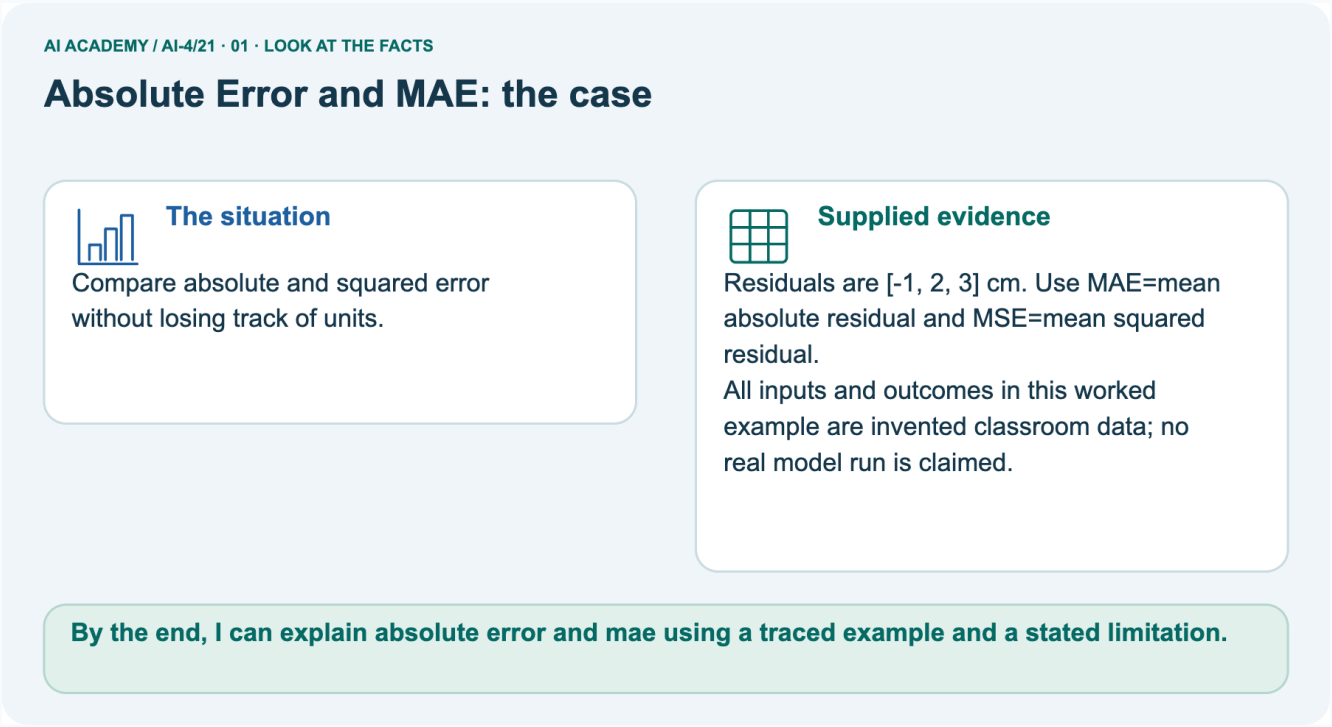
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Absolute Error and MAE graphic using its labels, deciding evidence and relevant limitation.

01. Absolute Error and MAE: the case



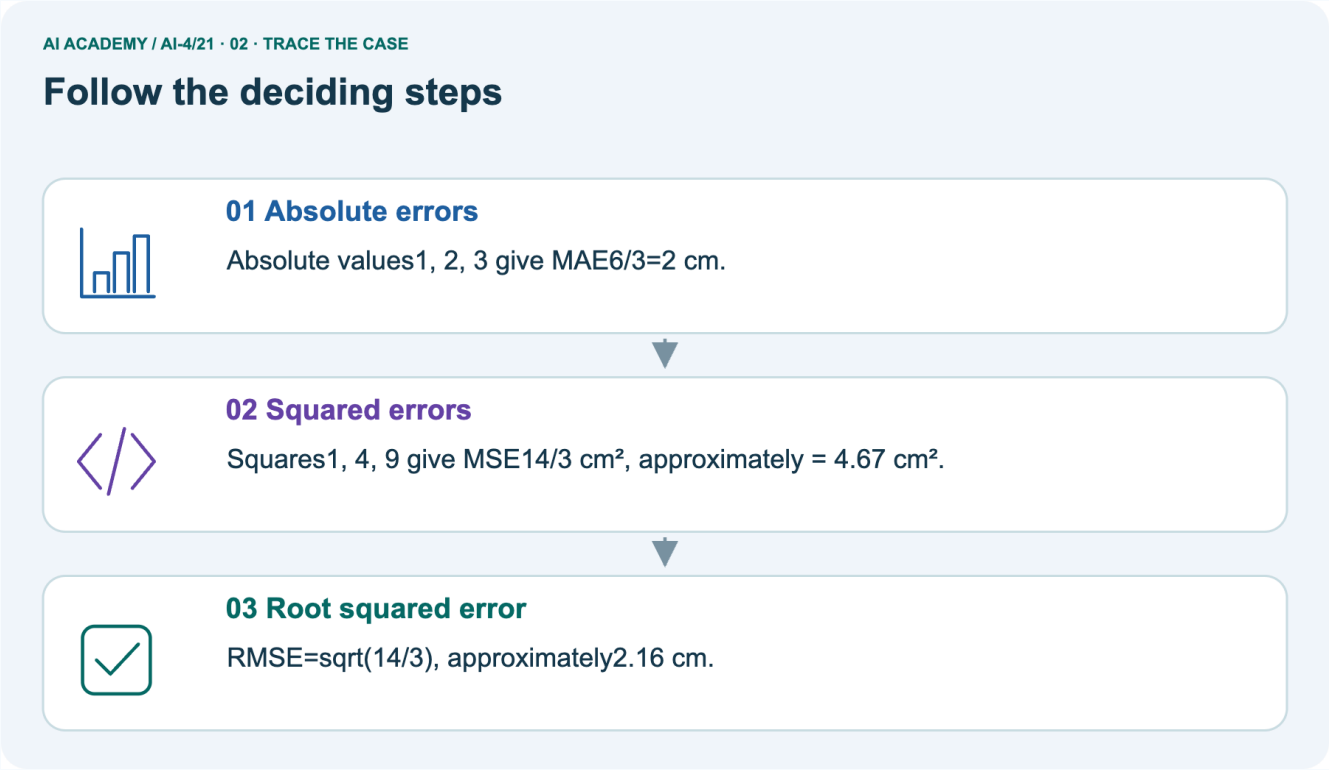
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

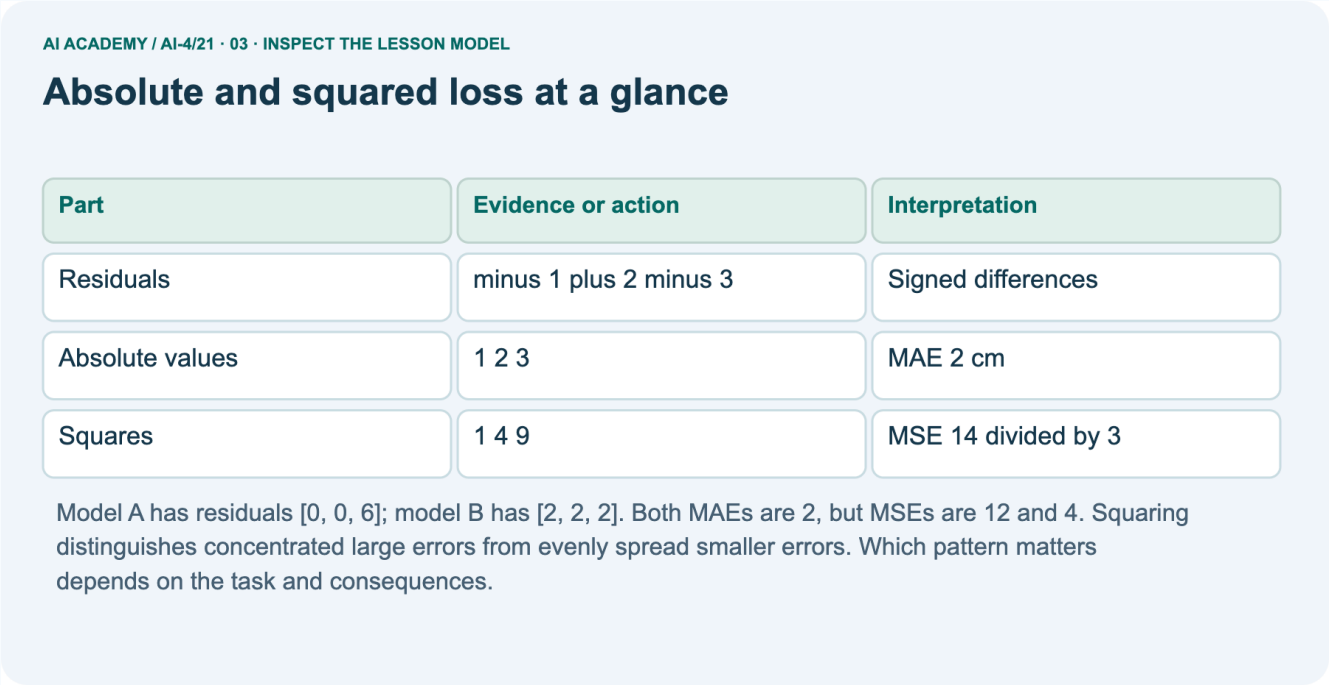


Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute errors: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Absolute and squared loss at a glance

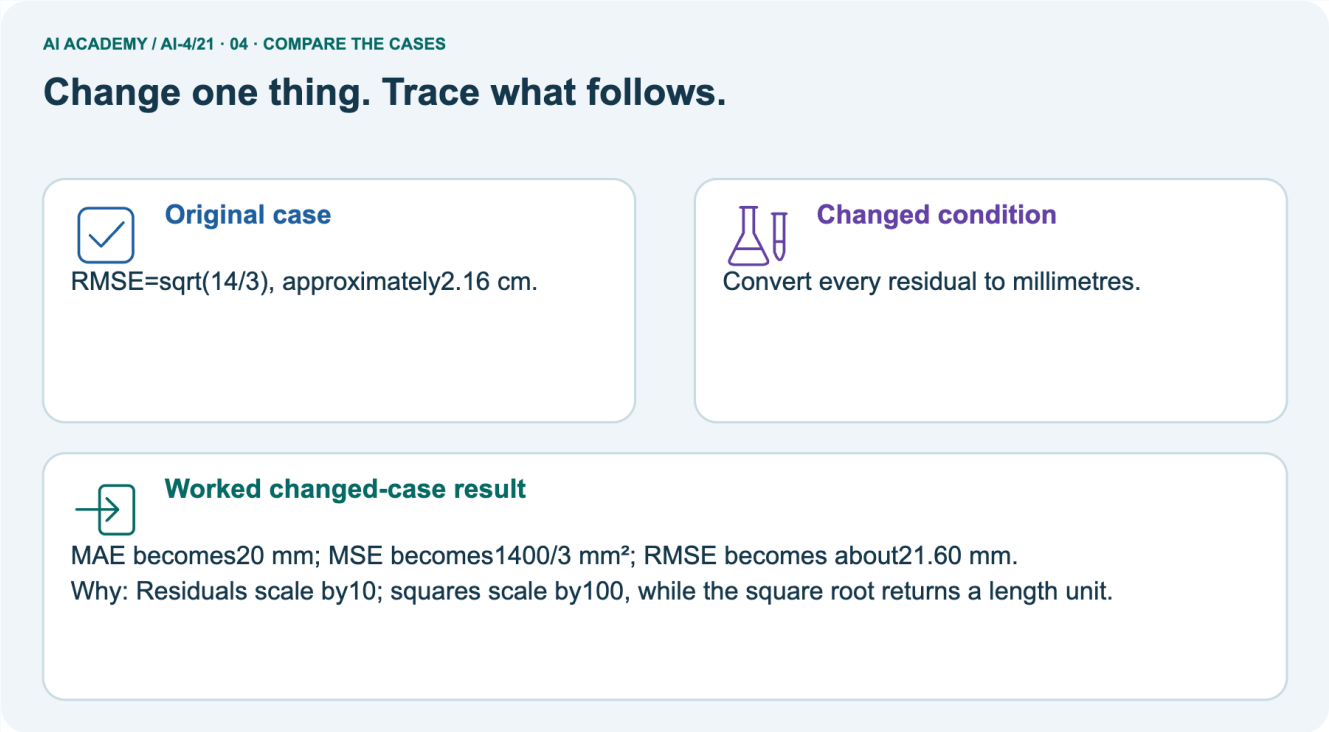


Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



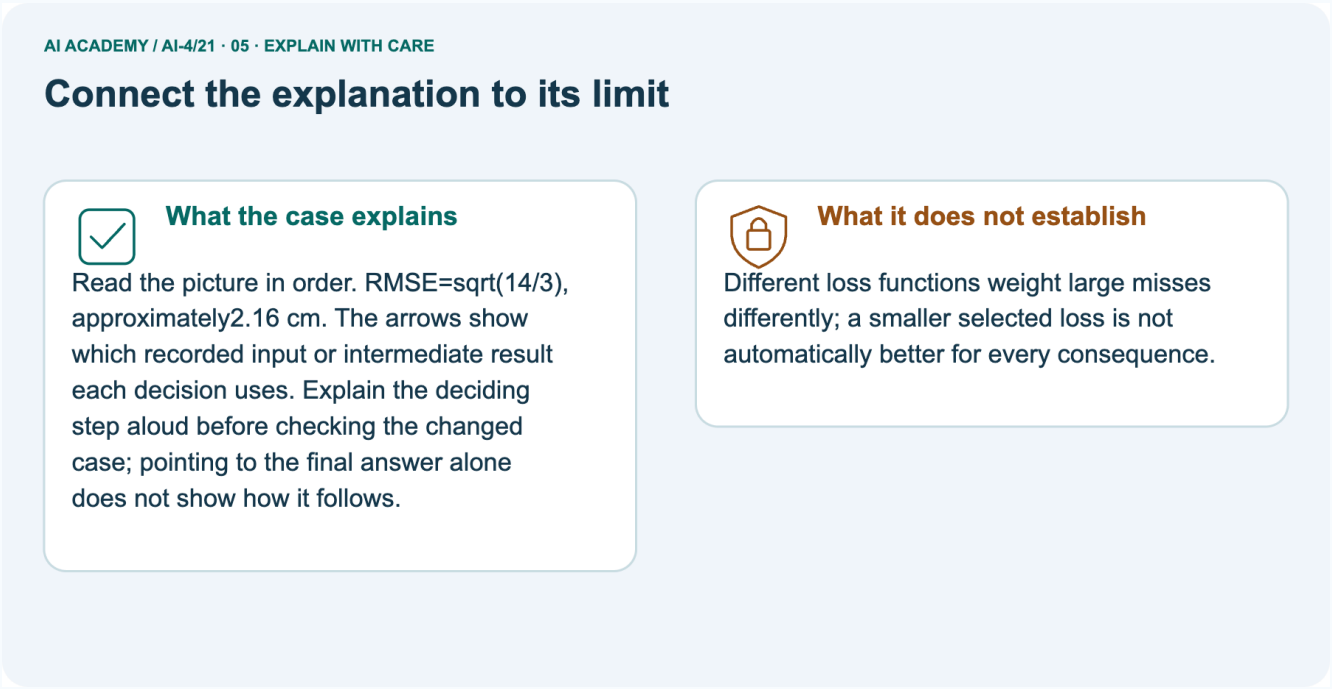
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

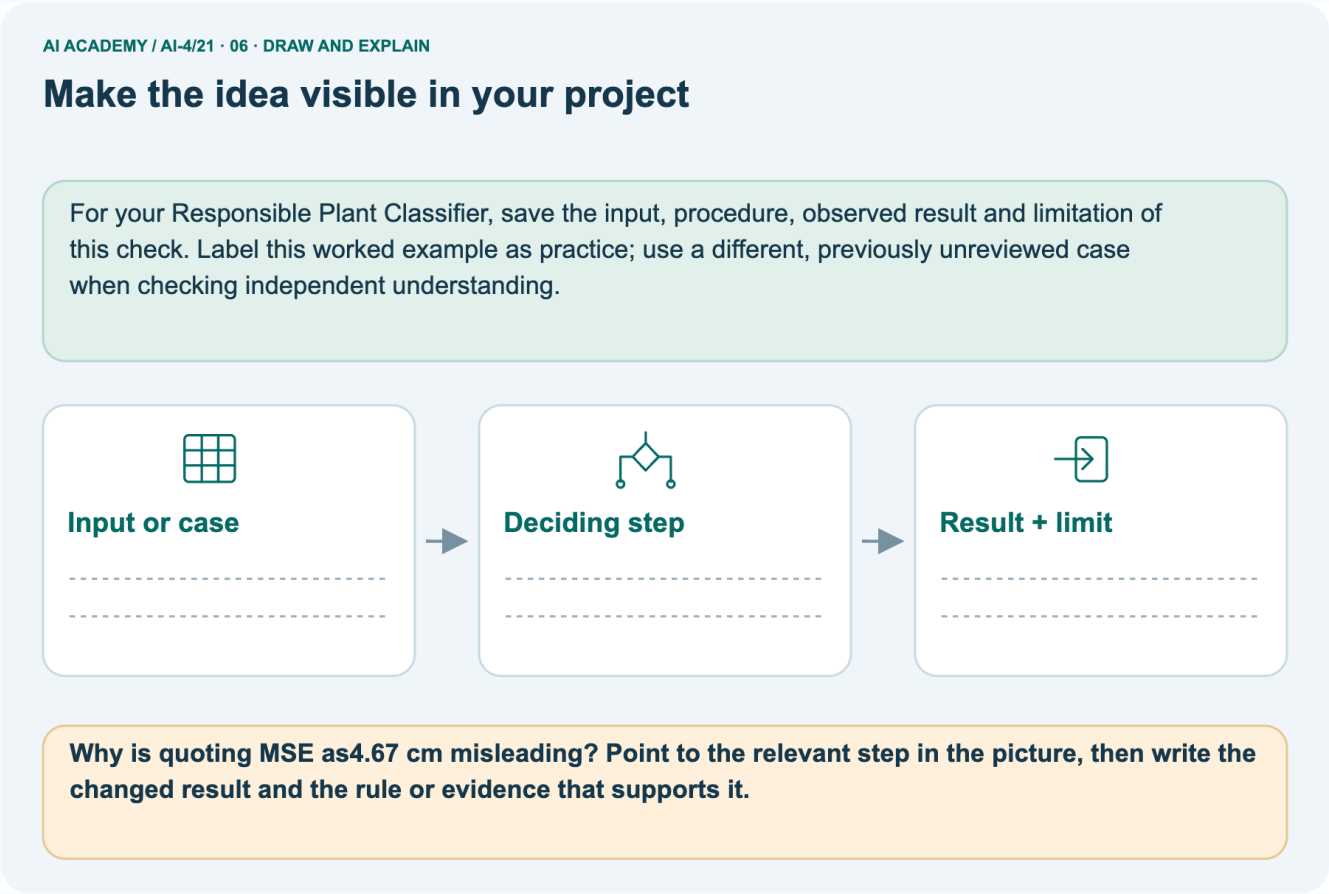
Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.